QUAID-I-AZAM UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCES

28/29

MID Term Examination
Fall 2023

Course: CS-414 Artificial Intelligence **Time: 90 minutes**

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Question No 1. Categorize the following as regression or classification problem. Give reason. (4 marks)

a) Predicting Car speed from camera image installed on top of the camera.

Regression

The car speed is going to vary at every point the camera clicks an image, It is going to be a range of values making it continuous.

01

b) To determine whether a transaction of Rs. 100,456 is a fraud or not.

Classification

Either the transaction is fraud or not, has only 2 possibile outcomes. A positive and a negative making it discrete.

c) Predicting company's profit at the end of the fiscal year based on performance in first two quarters.

Regression

Profit of the company is going to be some value based on data of the past. Making it continuous

01

d) To determine whether a product review of 500 words is positive or negative based on the content of the review.

Classification

01

Product review will have 2 discrete outcomes either positive or negative.

(08)

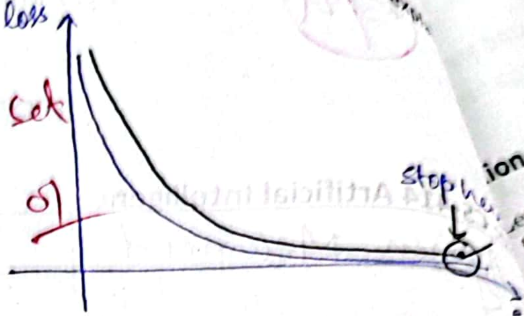
Question No 2. Give short answers. (2+2+2+3 marks)

a) How overfitting can be avoided using early stopping? Draw epoch vs loss graph to answer?

The point where we see that the loss has started to increase, we stop there.

on which data set & How?

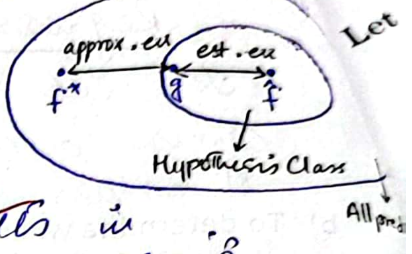
02



b) What is the difference between approximation and estimation error? How hypothesis class effects these errors?

Estimation Error → Difference between our predictor and the local minima.

Approximation Error → Difference between global minima and local minima of our Hypothesis Class

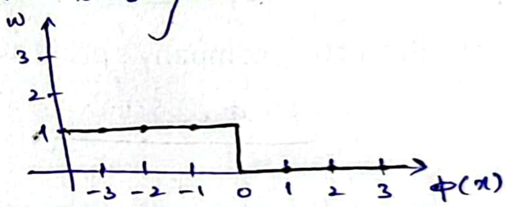


Increasing the size of hypothesis class ~~results~~ 02 in increasing the estimation error whereas decreasing the approximation error

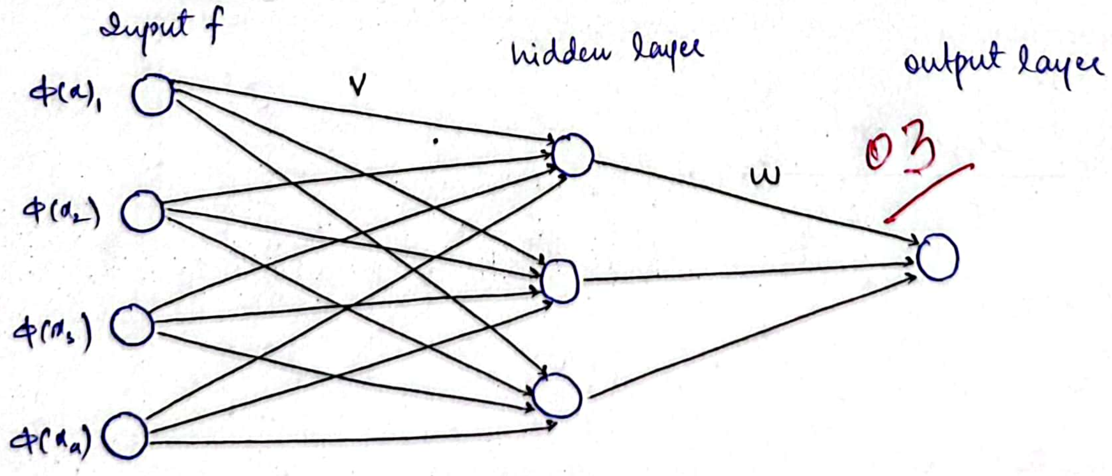
c) Can we use gradient descent algorithm with zero-one loss function? Justify your answer.

No, we cannot use GD algo. with 0-1 loss function because the gradient is zero almost everywhere.

02



d) Draw a neural network architecture with the following specifications. No. of input features = 4, units/neurons in hidden layer = 3, units/neurons in output layer = 1. Write down the dimensions of the learnable parameters V and W.



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Dimensions :-

V ⇒ (4×3)

W ⇒ (3×1)

$\phi(x)$
$\begin{bmatrix} 1, -2 \\ 1, -1 \end{bmatrix}$

08

$loss = (w \cdot \phi(x) - y)^2$

$(0, \frac{1}{2})(1, \ldots$

$0 - 1 = \ldots$

Question No 3. **(4 marks)**

i) Let $\phi(x) : \mathbb{R} \mapsto \mathbb{R}^d$, $\mathbf{w} \in \mathbb{R}^d$. Consider the following objective function (a.k.a. loss function).

$$\text{Loss}(x, y, \mathbf{w}) = \begin{cases} 1 - 2(\mathbf{w} \cdot \phi(x))y & \text{if } (\mathbf{w} \cdot \phi(x))y \leq 0 \\ (1 - (\mathbf{w} \cdot \phi(x))y)^2 & \text{if } 0 < (\mathbf{w} \cdot \phi(x))y \leq 1 \\ 0 & \text{if } (\mathbf{w} \cdot \phi(x))y > 1. \end{cases}$$

Let $d = 2$ and $\phi(x) = [1, x]$. Consider the following loss function.

$$\text{TrainLoss}(\mathbf{w}) = \frac{1}{2}\left(\text{Loss}(x_1, y_1, \mathbf{w}) + \text{Loss}(x_2, y_2, \mathbf{w})\right).$$

sq. loss

Compute TrainLoss(w) for the following values of (x1,y1) = (-2,1) ,(x2,y2) = (-1,-1) and w = [0, 1/2]

$\cancel{Loss(x_1, y_1, w) = (w \cdot \phi(x) - y)^2}$

04

$(w \cdot \phi(x_1)) * y_1 = [0, \frac{1}{2}][1, -2](+1) = 0 - 1(+1) = -1 \leq 0$

$(w \cdot \phi(x_2)) y_2 = [0, \frac{1}{2}][1, -1]\cancel{[-1]} = 0 - \frac{1}{2}(-1) = \frac{1}{2} = 0.5 \leq 1$

$Loss(x_1, y_1, w) = 1 - 2(w \cdot \phi(x))y$

$= 1 - 2([0, \frac{1}{2}][1, -2](1)) = 1 - 2(-1)$

$loss(x_2, y_2, w) = (1 - (w \cdot \phi(x))y)^2$

$= (1 - ([0, \frac{1}{2}][1, -1](-1)))^2 = (1 - 0.5$

$=$

$Trainloss(w) = \frac{1}{2}(Loss(x_1, y_1, w) + loss(x_2, y_2, w$

$= \frac{1}{2}(3 + 0.25)$

$= \frac{1}{2}(3.25)$

$\boxed{Trainloss(w) = 1.625}$

(4 marks)

(ii) Consider two-dimensional feature vector X = (x1,x2). Draw the line (i.e. the "decision boun... separates between vectors having a positive dot product with weights w = [3, −2] and those having... dot product. Shade the part of the 2D plane that contains vectors satisfying w · x > 0.

[Hint: It might help to write out the expression for the dot product and seeing the relation between... x2 that leads to a positive dot product. You could also use the geometric interpretation of the dot pro...

$X(x_1, x_2)$	w	$w \cdot X$
(0, 0)	(3, −2)	0
(−3, 1)	(3, −2)	−2 < 0
(−2, 1)	(3, −2)	−3 < 0
(−1, 1)	(3, −2)	0 → 0
(0, 1)	(3, −2)	1 > 0
(1, −1)	(3, −2)	0 → 0
(1, 1)	(3, −2)	2 > 0
(0, −1)	(3, −2)	−1 < 0
(2, 1)	(3, −2)	3 > 0
(3, 1)	(3, −2)	4 > 0
(1, 2)	(3, −2)	3 > 0
(−2, 3)	(3, −2)	1 > 0
(−1, 0)	(3, −2)	−1 < 0

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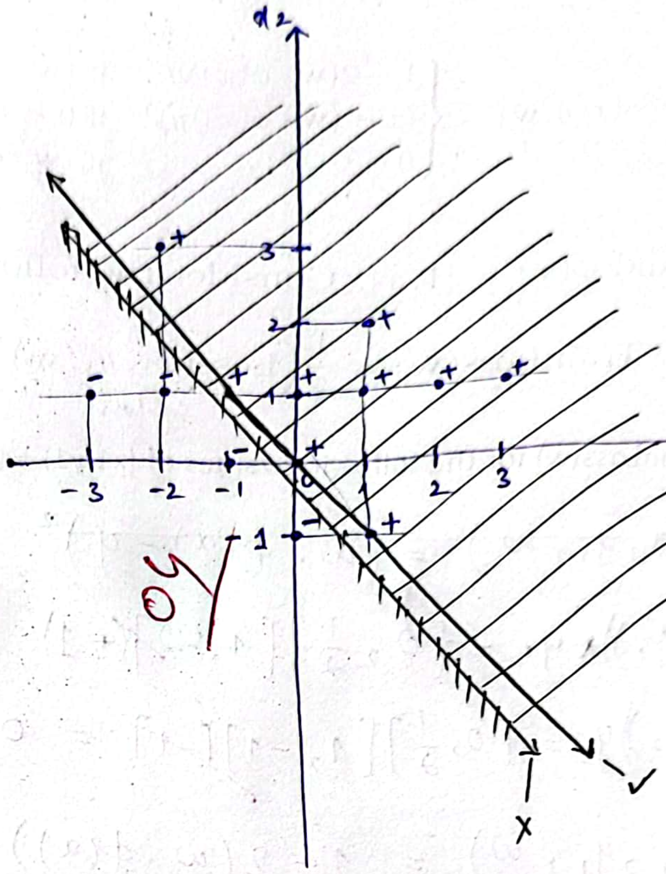

Question No. 5. Imagine you wish to recognize good and bad items produced by your company. You're able to measure three numeric properties of each widget: *P1, P2,* and *P3*. You randomly grab several items off of your shipping dock and extensively test whether or not they are good, obtaining the following results:

(Marks 4+4)

P1	*P2*	*P3*	*Result*
0	0	1	good
→ 9	1	2	bad
→ 5	0	3	good
→ 3	5	6	bad
2	0	4	good
6	5	0	? → bad.

Euclidean distance $= \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

L_1 Norm $= |x_2-x_1|^2 + |y_2-y_1|^2 + \dots$

(a). What a *three-nearest neighbor(3-NN)* algorithm would classify the following new example.
P1 = 6   *P2* = 5   *P3* = 0

$\sqrt{(6-0)^2+(5-0)^2+(0-1)^2} = \sqrt{6^2+5^2+1^2} = \sqrt{36+1+25} = \sqrt{62} = 7.87$

$\sqrt{(6-9)^2+(5-1)^2+(0-2)^2} = \sqrt{(-3)^2+4^2+(-2)^2} = \sqrt{9+16+4} = \sqrt{29} = \underline{5.38}$

$\sqrt{(6-5)^2+(5-0)^2+(0-3)^2} = \sqrt{1^2+5^2+(-3)^2} = \sqrt{1+25+9} = \sqrt{35} = \underline{5.91}$

$\sqrt{(6-3)^2+(5-5)^2+(0-6)^2} = \sqrt{3^2+(-6)^2} = \sqrt{9+36} = \sqrt{45} = \underline{6.70}$

$\sqrt{(6-2)^2+(5-0)^2+(0-4)^2} = \sqrt{4^2+5^2+(-4)^2} = \sqrt{16+25+16} = \sqrt{57} = 7.54$

So, the 3 nearest neighbours are (9, 1, 2), (5, 0, 3) and (3, 5, 6). Majority votes are for bad. Hence (6, 5, 0) will classified as <u>bad</u>.

...ming that we are not given the label i.e Result in above data set (part 1) and we wish to gro... their properties. Use K-means algorithm to find two clusters by taking initial values of Cen... (2, 0, 4). Run one iterations of the K-means algorithm.

(0, 0, 1), (9, 1, 2), (5, 0, 3), (3, 5, 6), (2...
(6, 5, 0) }

$\mu_1 = (0, 0, 1) \qquad \mu_2 = (2, 0, 4)$

$\sqrt{(0-0)^2+(0-0)^2+(1-1)^2} = 0 \longrightarrow 1$

$\sqrt{(2-0)^2+(0-0)^2+(4-1)^2} = 3.6$

$\sqrt{(0-9)^2+(0-1)^2+(1-2)^2} = 9.11$

$\sqrt{(2-9)^2+(0-1)^2+(4-2)^2} = 7.34 \longrightarrow C_2$

$\sqrt{(0-5)^2+(0-0)^2+(1-3)^2} = 5.38$

$\sqrt{(2-5)^2+(0-0)^2+(4-3)^2} = 3.16 \longrightarrow 2$

$\sqrt{(0-3)^2+(0-5)^2+(1-6)^2} = 7.68$

$\sqrt{(2-3)^2+(0-5)^2+(4-6)^2} = 5.47 \longrightarrow 2$

$= \sqrt{(0-2)^2+(0-0)^2+(1-4)^2} = 3.60$

$= \sqrt{(2-2)^2+(0-0)^2+(4-4)^2} = 0 \longrightarrow 2$

$= \sqrt{(0-6)^2+(0-5)^2+(1-0)^2} = 7.87$

$= \sqrt{(2-6)^2+(0-5)^2+(4-0)^2} = 7.54 \longrightarrow 2$

Cluster 1 : (0, 0, 1)

Cluster 2 : (9, 1, 2), (5, 0, 3), (3, 5, 6), (2, 0, ...

Centroids :-

<The End>

$\mu_1 = (0, 0, 1) \qquad \mu_2 = \left(\frac{25}{5}, \frac{11}{5}, \frac{15}{5}\right) = (5...$